Orbitra Platform Architecture

This document describes the production architecture of the Orbitra platform as of October 2026. It is maintained by the Platform team.

Services

Service	Language	Data store	Purpose
fleet-api	Go	PostgreSQL 16	Customer-facing REST API
telemetry-ingest	Rust	Kafka, TimescaleDB	Vehicle telemetry pipeline
ops-console	TypeScript (React)	-	Dispatcher web application
route-planner	Python	Redis	Route optimization workers
auth-service	Keycloak	PostgreSQL 16	SSO and API tokens

Data flow

Vehicles send telemetry every 5 seconds to telemetry-ingest, which writes raw events to Kafka and stores 1-minute rollups in TimescaleDB for 13 months. The fleet-api reads rollups for dashboards and raises alerts through the notification queue.

Reliability targets

- fleet-api availability SLO: 99.95% per month.
- Telemetry end-to-end latency: p95 under 3 seconds.
- Recovery point objective (RPO): 15 minutes; recovery time objective (RTO): 1 hour.